Repairing Geometric Brownian Motion: A Common PIT-Residual Test of the Merton, Heston, Variance Gamma, and Bates Models

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October 2026

Abstract

Geometric Brownian Motion (GBM) fails to describe S&P 500 returns in three ways: fat tails, volatility clustering, and persistent volatility shocks. This paper examines four established extensions of GBM (Merton jump diffusion, Heston stochastic volatility, Variance Gamma, and Bates stochastic volatility with jumps) and tests each on the same 2,519 daily S&P 500 log returns (September 2016–September 2026). Because these models do not assume Gaussian returns, they are compared on a common scale using probability integral transform (PIT) residuals, which are i.i.d. $N(0,1)$ under correct specification, with Monte Carlo simulation used for level statistics. The jump models (Merton, Variance Gamma) largely repair the shape of returns, cutting residual kurtosis from 19.7 to below 4, but leave volatility clustering untouched. Heston repairs most of the dynamics and has the best BIC, though it retains negative skew and mean-reverts too quickly. Bates improves the tails further at the cost of more residual clustering, and its jump component is weakly identified. No extension reproduces the full persistence of volatility observed in the data.

1 Extensions Concerning Geometric Brownian Motion

1.1 Jump Diffusion

"Option Pricing When Underlying Stock Returns are Discontinuous" Robert C. Merton [Mer76] The first extension this paper will concern itself with is Jump Diffusion, an idea proposed in the paper above by Robert C Merton, ands defined as the following: Total change in stock price:

1. Normal vibrations i.e., drift + variance
2. Abnormal vibrations (asymptote information arrival)

Hence... Jump Diffusion with a jump component modeled as a Poisson Process.
Poisson Distributed event is the arrival of important information about a/some stock.

- then arrivals are independently distributed
- then the probability of a jump formalizes as follows:
  Prob(the event about occurs $(t, t+h]) = \lambda h + O(h)$
  Prob(does occur $(t, t+h]) = 1 - \lambda h + O(h)$
  Prob(does occur more than one $(t, t+h]) = O(h)$
- λ = mean of # of arrivals per unit time
- $O(h) = \psi(h)$ if $\lim_{h\to 0} \frac{\psi(h)}{h} = 0$

This cumulative idea is modeled in the SDE:

$$\frac{dS}{S} = (\alpha - \lambda k)dt + \sigma dZ + dq$$

where α is the expected return, σ^2 is the variance of the return conditional on the Poisson process not occurring, dZ is a Wiener process, dq is the independent Poisson process with intensity λ, $k = E[Y-1]$ where $(Y-1)$ is the mean number of arrivals, $Y =$ change in stock price $= \frac{S_t}{S_{t-}}$, E is the expectation on Y.

Then from when modeled with the Poisson processes are as follows!

$$\frac{dS_t}{S_t} = (\alpha - \lambda k)dt + \sigma dZ \quad \text{if no Poisson process occurs}$$

$$\frac{dS_t}{S_t} = (\alpha - \lambda k)dt + \sigma dZ + (Y-1) \quad \text{if a Poisson process occurs}$$

with α, λ, k and σ as constants, a function of time t is:

$$S(t)/S = \exp\left[\left(\alpha - \frac{\sigma^2}{2} - \lambda k\right)t + \sigma Z(t)\right]\prod_{i=1}^{N_t} Y_i$$

α : expected return

σ : volatility

λ : intensity of Poisson process

k : $E[Y-1]$

$Z(t)$: Wiener or Wiener-process

Y_i : is the product of jump amplitudes to the i-th jump

The following is an explanation of why the above modeled as SDE forms the closed solution outlined above as well.

This explanation operates under the following assumptions: 1. Fitted probability space where Ω is the set of all possible outcomes; $\mathcal{F}$ is the collection of Ω (events), $\mathcal{F}_t$ is next-time (meaning $\mathcal{F}_t$ represents the history of information up to time t); $\mathbb{P}$ is a probability measure that assigns a number between $0 \leq \mathbb{P} \leq 1$ to every event in $\mathcal{F}$ where $\mathbb{P}(\Omega) = 1$. We will assume usual conditions (to contain all $\mathbb{P}$-null sets of $\mathcal{F}_t = \cap_{s>t}\mathcal{F}_s$) or filtration is right-continuous.

The uncertainty within this system is driven by 3 independent random variables: 1. Standard Brownian Motion (Z_t) 2. Poisson Process (N_t w/ rate λ) 3. Jump Magnitudes $(Y_i > 0)$

It may also be useful to note $\lim_{s\to t^-} S_s = S_{t-}$ represents the stock price the instant prior to a potential jump at time t.

Putting this formally we see:

$$\frac{dS_t}{S_{t-}} = (\alpha - \lambda k)dt + \sigma dZ_t + d\left(\sum_{i=1}^{N_t}(Y_i - 1)\right)$$

where dq is the same as $d\left(\sum_{i=1}^{N_t}(Y_i-1)\right)$ and where $k = E[Y_i - 1]$ is the expected relative jump size.

Here we must note since the S_t process, at least one at a certain time t has sudden jumps, the standard (continuous) Itô's Lemma doesn't apply within this system, we work with a semimartingale formulation of Itô's Lemma for non-continuous processes as outlined below:

on $f(S_t) = \ln(S_t)$ we see

$$d\ln(S_t) = f'(S_{t-})dS_t^c + \frac{1}{2}f''(S_{t-})d\langle S^c, S^c\rangle_t + [\ln(S_t) - \ln(S_{t-})]$$

Evaluating the first term within our specific SDE (Sec...) we see

$$f'(S_{t-})dS_t^c = \frac{1}{S_{t-}}\left((\alpha - \lambda k)S_{t-}dt + \sigma S_{t-}dZ_t\right) = (\alpha - \lambda k)dt + \sigma dZ_t$$

Evaluating the second term we see...

$$d\langle S^c, S^c\rangle_t = \sigma^2 S_{t-}^2 dt \implies \frac{1}{2}\left(\frac{-1}{S_{t-}^2}\right)(\sigma^2 S_{t-}^2 dt) = -\frac{1}{2}\sigma^2 dt$$

Combining these terms we obtain the continuous section of the following:

$$\left(\alpha - \frac{1}{2}\sigma^2 - \lambda k\right) dt + \sigma dZ_t$$

Now when a jump occurs at a time t_i, the price jumps from S_{t-} to S_{t_i} where $Y_i = S_{t_i}/S_{t-}$ and calculating the log-difference we see

$$\ln(S_{t_i}) - \ln(S_{t-}) = \ln\left(\frac{Y_i S_{t-}}{S_{t-}}\right) = \ln(Y_i)$$

and summing over N_t we see

$$\sum_{i=1}^{N_t} \ln(Y_i)$$

Combining all these elements we realize our original SDE of

$$d\ln(S_t) = \left(\alpha - \frac{1}{2}\sigma^2 - \lambda k\right) dt + \sigma dZ_t + \ln(Y_{N_t}) dN_t$$

Integrating and exponentially (See appendix for full algebraic simplification) we resolve:

$$S_t = S_0 \exp\left[\left(\alpha - \frac{1}{2}\sigma^2 - \lambda k\right) t + \sigma Z_t\right] \prod_{i=1}^{N_t} Y_i$$

In summary, you can see how this idea is quite useful in pricing assets such as european call options that often suffer from price discontinuity or "jumps". Another alternative/extension to GBM is outlined below: –please note that these alternatives are simply outlined and formally introduced here and will be rigourosly tested and analyzed in the next section (see Section 6)

1.2 Heston's Model

Under the risk-neutral pricing measure $\mathbb{Q}$, the asset price process S_t and its variance process v_t evolve via coupled Stochastic Differential Equations (SDEs)[1]:

Key Equations

$$dS_t = rS_t\,dt + \sqrt{v_t} S_t\,dW_t^S \tag{1}$$

$$dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v \tag{2}$$

$$d\langle W^S, W^v\rangle_t = \rho\,dt, \quad \rho \in [-1, 1] \tag{3}$$

Component Breakdown

1. Asset Price SDE

- $rS_t\,dt$: Risk-free rate drift component
- $\sqrt{v_t}$: Stochastic volatility factor (varies dynamically over time t)
- dW_t^S: Standard Brownian motion driving asset price shocks

[1]The model is due to [Hes93].

2. Variance SDE

- θ: Long-run equilibrium variance
- κ: Speed of mean reversion ($\kappa > 0$)
- ξ: Volatility of volatility (vol-of-vol, $\xi > 0$)
- $\sqrt{v_t}$: Cox-Ingersoll-Ross (CIR) square-root diffusion parameter (ensures $v_t \geq 0$)

3. Correlation Structure

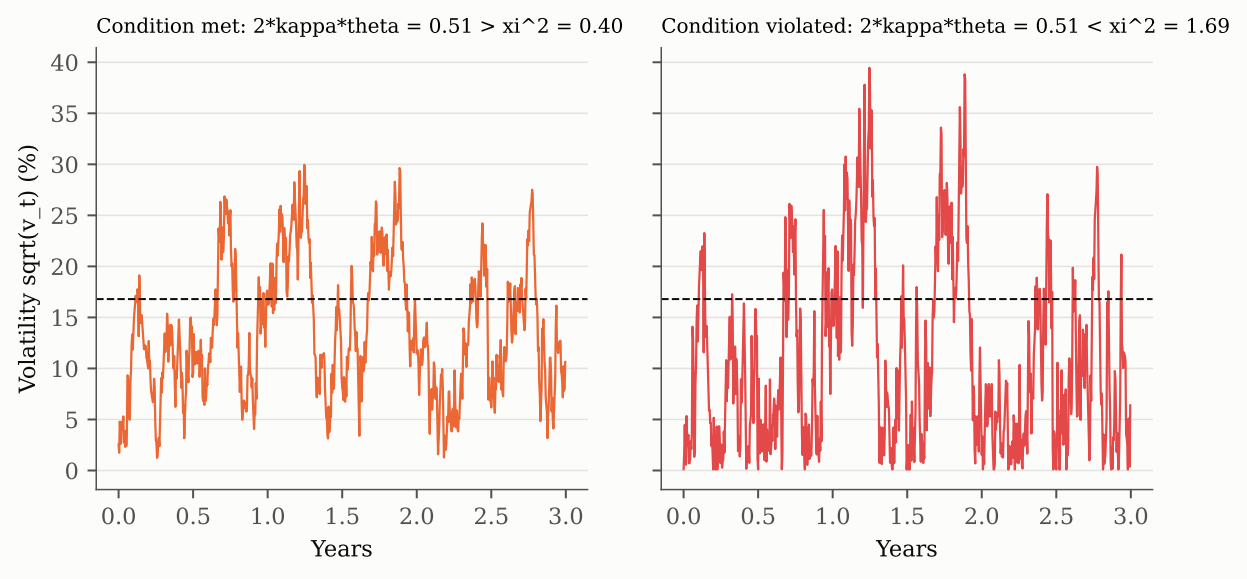
- $\rho = \text{corr}(dW_t^S, dW_t^v)$: Instantaneous correlation between asset and variance shocks
- $\rho < 0$: Typical for equity markets (leverage effect: price drops correlate with volatility spikes)
- $\rho > 0$: Typical for certain commodity markets

Feller Condition

To guarantee that the variance process v_t remains strictly positive ($v_t > 0$) almost surely and never hits zero, the model parameters must satisfy:

$$2\kappa\theta > \xi^2 \tag{4}$$

where the standard parameter domain restrictions are $\kappa > 0$, $\theta > 0$, and $\xi > 0$.



Log-Price Dynamics and Analytical Derivations

To apply this model better and to be able to test it more rigorously, we apply the following $f(\cdot)$ to the equations above.

$f(S_t) = \ln(S_t)$:

$$d\ln(S_t) = f'(S_t)\, dS_t + \frac{1}{2} f''(S_t)\, d\langle S, S\rangle_t \tag{5}$$

with the first term:

$$f'(S_t)\, dS_t = \frac{1}{S_t}\left(rS_t\, dt + \sqrt{v_t} S_t\, dW_t^S\right) = r\, dt + \sqrt{v_t}\, dW_t^S \tag{6}$$

with a second term of:

$$d\langle S,S\rangle_t = (\sqrt{v_t}S_t)^2\,dt = v_t S_t^2\,dt \tag{7}$$

$$\frac{1}{2}f''(S_t)\,d\langle S,S\rangle_t = \frac{1}{2}\left(-\frac{1}{S_t^2}\right)(v_t S_t^2\,dt) = -\frac{1}{2}v_t\,dt \tag{8}$$

Combining these we synthesize:

$$d\ln(S_t) = \left(r - \frac{1}{2}v_t\right)dt + \sqrt{v_t}\,dW_t^S \tag{9}$$

Integrating both these terms we get as follows:

$$S_t = S_0 \exp\left[\int_0^t \left(r - \frac{1}{2}v_s\right)ds + \int_0^t \sqrt{v_s}\,dW_s^S\right] \tag{10}$$

$$v_t = v_0 + \kappa\theta t - \kappa\int_0^t v_s\,ds + \xi\int_0^t \sqrt{v_s}\,dW_s^v \tag{11}$$

It is also interesting to note that the Feller condition is derived via applying Itô's lemma to $f(v_t) = \ln(v_t)$ on the variance SDE, outlined below:

$$d\ln(v_t) = \left[\frac{\kappa\theta - \frac{1}{2}\xi^2}{v_t} - \kappa\right]dt + \frac{\xi}{\sqrt{v_t}}\,dW_t^v \tag{12}$$

Right away you see it is evident that the drift term's numerator $\left(\kappa\theta - \frac{1}{2}\xi^2\right)$ is the same form as the Feller condition (obviously without the inequality) and here is where it derives its importance as $2\kappa\theta > \xi^2$ as this guarantees the drift (which is modeled here as a $2\kappa\theta > \xi^2$) tends to pos. infinity $(+\infty)$ creating a force that makes $(v_t > 0$ strictly).

This model offers us the idea that volality is not constant and is modeled as a seperate random process. This is model does not involve jumps but the speed and spread of provement do both constantly expand and contract. It also, using the third equation outlined in the Componment Breakdown subsection, leverages the effect of asset price to volatility, i.e. holding that when an asset price drops, the volatility of that asset tends to spike. This will be further tested in the next section as well.

1.3 Variance Gamma Distributions

The Variance Gamma process of modeling asset prices is labeled as a "pure-jump stochastic process"

Meaning unlike Merton or Heston, V-Gamma processes eliminate Brownian motion entirely. But instead models asset prices using an infinite number of small jumps in every time interval (finite) meaning it is an infinite activity process.[2]

A standard geometric Brownian motion process is defined as:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

V-Gamma takes a standard GBM and replaces deterministic time t with a random economic time subordinator (G_t), so it is defined as below:

$$S_t = S_0 \exp\left(rt + \omega t + X(G_t;\theta,\sigma)\right)$$

where G_t follows a Gamma process w/ a mean rate $= 1$ & variance ν

The VG Process evaluated at a deterministic time t is

$$X(\theta,\sigma,\nu) = \theta G_t + \sigma W(G_t)$$

where W_t is a Wiener process independent of G_t & $G_t \sim \text{Gamma}\left(\frac{t}{\nu},\nu\right)$

In this process:

[2]The Variance Gamma model is due to [MS90] and [MCC98].

- σ : controls volatility of the Wiener Process
- θ : controls skewness
- ν : controls Kurtosis (variance rate of economic time process)
- G_t : Gamma process ($\uparrow \nu$ means more clustering / activity)

This process also has several key mathematical properties:

- **Pure jump** : X_t has no Gaussian path component
- **Infinite Activity** : X_t is infinitely "jumping" while having a finite variation

* the V-G process can essentially be written as the difference in 2 independent Gamma process:

$$X_t = \Gamma_t^+ - \Gamma_t^-$$

where Γ_t^+ represents an upward jump & Γ_t^- represents a downward jump

1.4 Bates Model

This is the final extension this paper will concern itself with. This model, proposed by David Bates in 1996 [Bat96], essentially combines Heston & Merton's models.

Bates begins like Heston, with 2 SDEs under a risk-neutral measure $\mathbb{Q}$, asset price S_t, and variance v_t:

$$\begin{aligned} dS_t &= (r - q - \lambda\bar{k})S_t\,dt + \sqrt{v_t}S_t\,dW_t^S + (J-1)S_t\,dN_t \\ dv_t &= \kappa(\theta - v_t)\,dt + \sigma\sqrt{v_t}\,dW_t^v \end{aligned}$$

where:

- r, q are risk-free rate & dividend yield respectively.
- v_t is the instantaneous variance of the asset return.
- dW_t^S, dW_t^v are Wiener processes under $\mathbb{Q}$, correlated w/ ρ formulated as:

$$d\langle W^S, W^v\rangle_t = \rho\,dt$$

- κ is the speed of mean reversion for variance.
- θ is the long-term mean of variance.
- σ is the volatility of the volatility.
- dN_t is a Poisson process w/ a constant jump force of λ (which is a force independent of the Wiener processes).
- J is the jump size/magnitude, w/ a mean of $1 + \bar{k}$ so that if some jump occurs at time t, $S_t = S_{t^-} \cdot J$.
- $\bar{k}$ is the expected percentage jump size used to fix the drift so that $\mathbb{E}\left[\frac{dS_t}{S_t}\right] = (r-q)\,dt$, and for $\bar{k}$, following a log-normal jump assumption, $\ln(J) \sim \mathcal{N}\left(\mu_J, \sigma_J^2\right)$ so that:

$$\bar{k} = \exp\left[\mu_J + \frac{1}{2}\sigma_J^2\right] - 1$$

If you've at all familiar with this model, you'll already see a possible issue, as I specified without λ tract parameters, non-robust the model's PIDE is heavy on the Merton model, a PIDE or partial integro-differential equation must be used due to the jumping Poisson process.

Applying Itô's Lemma for jump diffusion process:

$$\frac{\partial V}{\partial t} + \frac{1}{2}vS^2\frac{\partial^2 V}{\partial S^2} + \rho\sigma vS\frac{\partial^2 V}{\partial S \partial v} + \frac{1}{2}\sigma^2 v\frac{\partial^2 V}{\partial v^2} + (r - q - \lambda\bar{k})S\frac{\partial V}{\partial S} + \kappa(\theta - v)\frac{\partial V}{\partial v} - rV + \lambda\mathbb{I}[V] = 0$$

under an European option price $V(S, v, t)$ when the integral term ($\mathbb{I}[V]$) captures non-local jumps as:

$$\mathbb{I}[V] = \int_0^\infty \left[V(J \cdot S, v, t) - V(S, v, t) - (J - 1)S\frac{\partial V}{\partial S}\right] dJ$$

Since Bates model is structured as a log-normal jump distribution (affine class), and therefore its closed-form solution operates under $x_t = \ln(S_t/S_0)$.

Since this model is affine, i.e., (x_t, v_t) drift, Var-Covar matrix, and jump arrival intensity are linear functions of (x_t, v_t).

Since this has been proved [DPS00], the Feynman-Kac theorem guarantees that conditional expectation can take an exponential function form:

$$\phi(u, \tau, T) = \exp\left(C(\tau, u) + D(\tau, u)v_t + iux_t\right)$$

expanded when

If $\phi(u, \tau)$ is the conditional characteristic function of x_T in state (v_t, S_t) at $t = T - \tau$ then:

$$\phi(u, \tau) = \mathbb{E}^{\mathbb{Q}}\left[e^{iux_T} \mid x_t, v_t\right] = \exp\left(C(\tau, u) + D(\tau, u)v_t + iux_t\right)$$

Notice remarkably, since the jump component N_t is independent of the Heston process (w_t^S, w_t^v), the characteristic factorizes as:

$$\phi_{\text{Bates}}(u, \tau) = \phi_{\text{Heston}}(u, \tau) \cdot \phi_{\text{Jump}}(u, \tau)$$

Now, while this factorizes neatly, it's important to expand upon the intuition behind this. They are as such: Drawing from Feynman-Kac theorem of exponential affine-functional form, and under the law of iterated expectations (the raw expectation ϕ must be a martingale under risk-neutral measure $\mathbb{Q}$). By applying Itô's Lemma to $\phi(u, \tau)$, PIDE into a system of Riccati equations we realize the process drift is zero, and further converting the original process $C(\tau, u) + D(\tau, u)$. Where $D(\tau, u)$ governs variance dependency and $C(\tau, u)$ governs deterministic time accumulation with boundary conditions $D(0, u) = 0$, $C(0, u) = 0$ respectively.

Formulated we realize the following ordinary differential equation:

$$\begin{aligned}
\frac{\partial C}{\partial \tau} &= (r - q)iu + \kappa\theta D(\tau, u) + \lambda\mathbb{E}\left[e^{iu\ln J} - 1\right] - \lambda\bar{k}iu \\
\frac{\partial D}{\partial \tau} &= \frac{1}{2}\sigma^2 D^2 + (\rho\sigma iu - \kappa)D - \frac{1}{2}(u^2 + iu)
\end{aligned}$$

Since we clearly see the jump term (λ) is independent of v_t and since $C(\tau, u)$ integration is a linear process, it can yield two independent integrals:

$$C(\tau, u) = C_{\text{Heston}}(\tau, u) + C_{\text{Jump}}(\tau, u)$$

Exponentiating, we can factorize back to the form:

$$\phi_{\text{Bates}}(u, \tau) = \exp\left(iux_0 + D(\tau, u)v_0 + C_{\text{Heston}}(\tau, u)\right) \cdot \exp\left[C_{\text{Jump}}(\tau, u)\right]$$

which yields:

$$\phi_{\text{Bates}}(u, \tau) = \phi_{\text{Heston}}(u, \tau) \cdot \phi_{\text{Jump}}(u, \tau)$$

Evaluating these two components we get:

1. Heston Evaluation

$$D(\tau, u) = \frac{\kappa - \rho\sigma i u - d}{\sigma^2}\left(\frac{1 - e^{-d\tau}}{1 - ge^{-d\tau}}\right)$$

Integrating this $D(\tau, u)$ expression we realize $C_{\text{Heston}}(\tau, u)$ expressed as:

$$C_{\text{Heston}}(\tau, u) = (r - q)iu\tau + \frac{\kappa\theta}{\sigma^2}\left[(\kappa - \rho\sigma i u - d)\tau - 2\ln\left(\frac{1 - ge^{-d\tau}}{1 - g}\right)\right]$$

where we meet with $d = \left[(\rho\sigma i u - \kappa)^2 + \sigma^2(iu + u^2)\right]^{1/2}$ and where $g = \frac{\kappa - \rho\sigma i u - d}{\kappa - \rho\sigma i u + d}$.

Note that branch gets or beyond the scope return for context which is included in the appendix of a covered example.

*2. Merton Jump Evaluation

Since log jump size $\ln(J) \sim \mathcal{N}\left(\mu_J, \sigma_J^2\right)$, it a moment generating function in its closed form is:

$$\mathbb{E}\left[e^{iuY}\right] = \exp\left(iu\mu_J - \frac{1}{2}\sigma_J^2 u^2\right)$$

Now, integrating $C(\tau, u)$ for C_{Jump} directly over the closed interval from $0, \tau$ we see:

$$C_{\text{Jump}}(\tau, u) = \lambda\tau\left(e^{iu\mu_J - \frac{1}{2}\sigma_J^2 u^2} - 1 - iu\bar{k}\right)$$

$$C_{\text{Jump}}(\tau, u) = \lambda\tau\left[\exp\left(iu\mu_J - \frac{1}{2}\sigma_J^2 u^2\right) - 1 - iu\bar{k}\right]$$

Finally yielding:

$$\phi_{\text{Jump}}(u, \tau) = \exp\left(C_{\text{Jump}}(u, \tau)\right)$$

where $\bar{k} = \exp\left(\mu_J + \frac{1}{2}\sigma_J^2\right) - 1$ presents double counting of the expected spot rate in an asset price.

While certainly a landmark in mathematical finance theory, the Bates model does seem to have several issues.

2 Empirical Results Concerning GBM Extensions

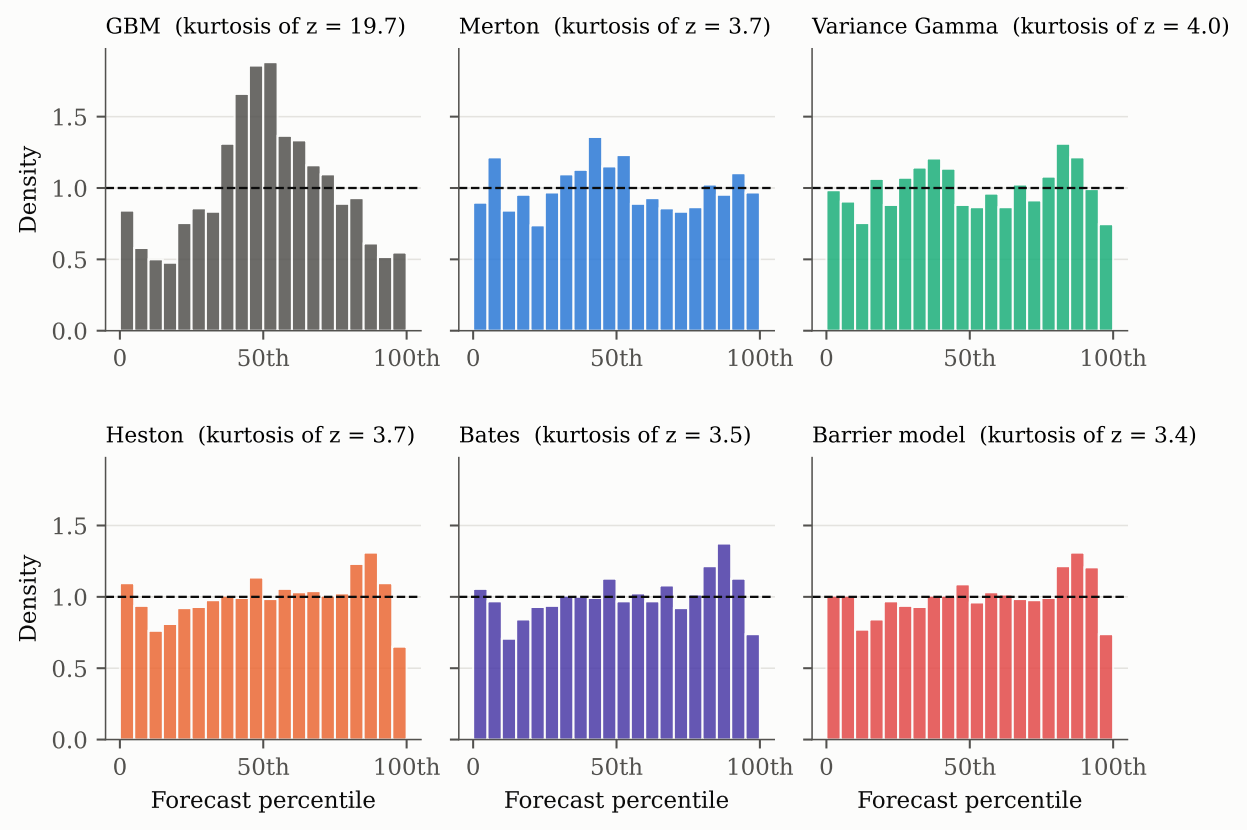
2.1 Adjustments of the Testing Framework

In order to rigourosuly test these four above models, we will employ the same framework outlined in Section 3 (Statistical Methodology) with some adjustments to better apply this papers framework to these methods. It is important to note several key changes before outlining the testable system. Firstly, it is clear that we cannot test these four models in an identical way to the way in which we tested GBM. GBM is tested under the null assumption that returns will display Gaussian properties. These new models do not operate with that null assumption, as they are extensions of GBM designed to model the non-normal log returns GBM failed to. Therefore, we need a new system. This is where we will PIT (Probability Integral Transform) residuals [Ros52, DGT98]. Operating under a forecast distribution modeled after F_t which is what we will notate this frameworks CDF that will formulate day t returns given all returns of day's $t - 1$. Using this we can formulate the following equations:

- $u_t = F_t(r_t)$ where r_t represents the returns at some time t
- $z_t = \Phi^{-1}(u_t)$

This serves to create the null conditions of these four models tests as if they predict returns accurately, then z_t will realize as $N(0, 1)$. This is because if the model's forecasts are correct, those ranks behave like independent random draws (flat, no clustering), which after a normal transform become i.i.d. N(0,1). Secondly, while the above derivations of the Heston and Bates models are written under a risk neutral measure Q in order to be in the correct form for options pricing, to test this framework we will reformulated these models under the probability measure P in order to more accurately test them against real market

data. This simply means that the $(r - q)$ term in these models will be replaced with a general drift term μ. All four of these models will be tested under the same window as the Section 4 test was done under, i.e 2519 daily log returns. Third, the ADF and OU tests are run on levels (log-price and rolling volatility) rather than daily returns, so the PIT does not apply to them. For these we simulate 1,000 paths of 2,519 days from each fitted model, compute the same statistic on each path, and report where the S&P 500 value falls in that simulated distribution (marked MC in the tables). The variance ratio on rolling volatility is treated the same way, using i.i.d. $N(0, 1)$ series as the null.



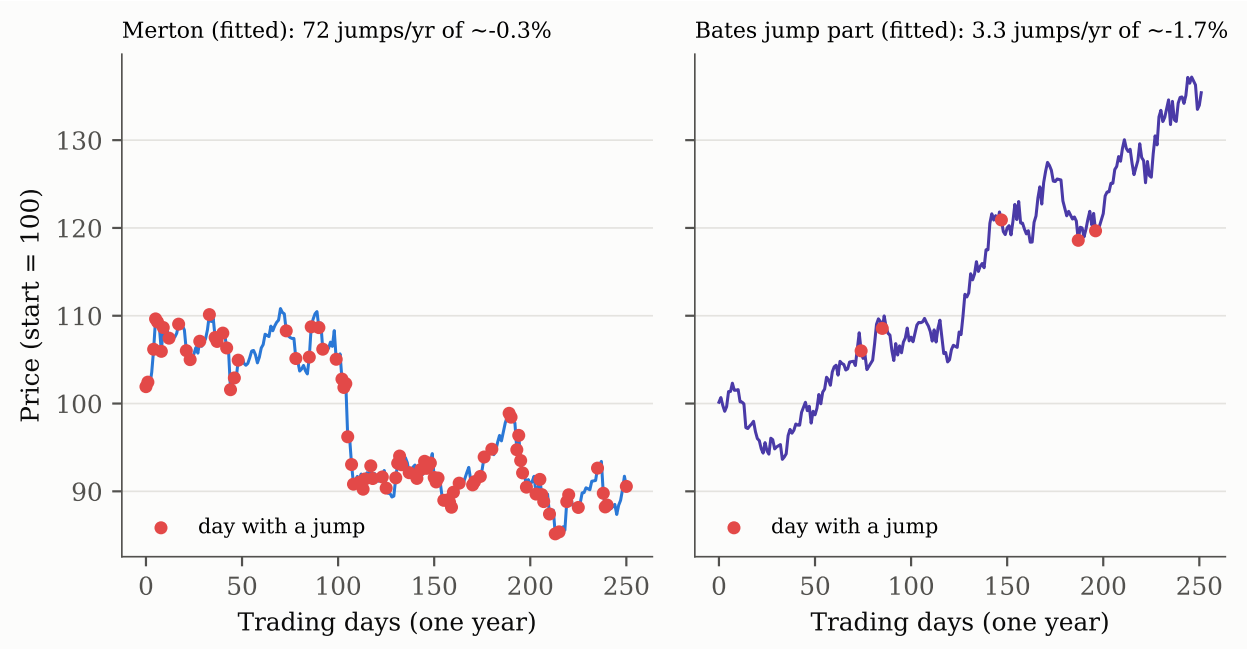
2.2 Jump Diffusion Model Results

Jumps correct the shape of returns: kurtosis of z_t falls from 19.69 to 3.73 and skewness to roughly zero, while the log-likelihood rises by 430. Yet volatility and jump intensity are constant, so Merton forecasts every day identically; squared-residual dependence is not reduced ($Q_{sq}(10) = 3{,}141.5$), and its implied volatility half-life (13.6 days) is indistinguishable from GBM's. The fitted 72 small jumps per year largely stand in for time-varying volatility rather than rare crashes.

Table 1: Specification diagnostics – Merton Jump Diffusion (S&P 500, 2016-09-01 to 2026-09-11, $N = 2{,}519$)

Statistic	GBM (Part 4)	Merton	Reference	p-value (Merton)
Panel A: Distribution of PIT residuals				
Skewness	−0.678	0.038	0	
Kurtosis	19.69	3.730	3	
Jarque-Bera	29,431.8	56.6	$\chi^2(2)$	<0.001
Panel B: Serial dependence (Ljung-Box)				
$Q_{\text{raw}}(10)$	255.7	59.3	$\chi^2(10)$	<0.001
$Q_{\text{raw}}(100)$	427.1	184.8	$\chi^2(100)$	<0.001
$Q_{\text{sq}}(10)$	2,741.2	3,141.5	$\chi^2(10)$	<0.001
$Q_{\text{sq}}(100)$	3,281.7	5,062.7	$\chi^2(100)$	<0.001
Panel C: Variance ratios				
VR(2), cumulated returns / z	0.856	0.929	1	0.014
VR(20), cumulated returns / z	0.806	0.735	1	0.055
VR(10), realized vol 21d	2.579	2.265	1	<0.001 (MC)
VR(21), realized vol 63d	5.458	5.613	1	<0.001 (MC)
Panel D: Level statistics, model-implied vs. S&P 500 (1,000 simulated paths)				
ADF, log price	−0.86	−0.97 [−2.44, 0.81]	−0.48 (S&P)	0.692 (MC)
ADF, realized vol 21d	−6.79	−6.73 [−7.62, −5.82]	−5.08 (S&P)	<0.001 (MC)
OU half-life, RV 21d (days)	13.2	13.6 [11.5, 16.1]	100.5 (S&P)	<0.001 (MC)
OU half-life, RV 63d (days)	40.8	41.0 [30.3, 56.4]	428.0 (S&P)	<0.001 (MC)
Panel E: Fit				
Log-likelihood	7,692.8	8,123.3		
AIC	−15,381.7	−16,236.6	lower is better	
BIC	−15,370.0	−16,207.4	lower is better	
Parameters (k)	2	5		

Notes: 'GBM (Part 4)' repeats the Section 4 statistic (raw returns are GBM's PIT residuals); 'Merton' is the same statistic on this model's PIT residuals $z_t = \Phi^{-1}(F_t(r_t))$, i.i.d. $N(0,1)$ under correct specification; 'Reference' is the value a correct model should produce. p-values refer to the Merton column: asymptotic, except (MC) = Monte Carlo (Panel C realized-vol rows against 2,000 i.i.d. $N(0,1)$ series; Panel D: S&P value against 1,000 paths simulated from the fitted model). Variance-ratio p-values use the heteroskedasticity-robust z^*. Panel D: GBM and model columns are simulated medians, model 90% band in brackets; OU p-values are computed on the AR(1) coefficient b. Estimates (annualised, under $\mathbb{P}$): $\alpha = 0.1412$, $\sigma = 0.09792$, $\lambda = 71.84$, $\mu_J = -0.002961$, $\sigma_J = 0.01667$. With ∼72 jumps per year of ∼ −0.3%, the jumps largely stand in for time-varying volatility rather than rare crashes.



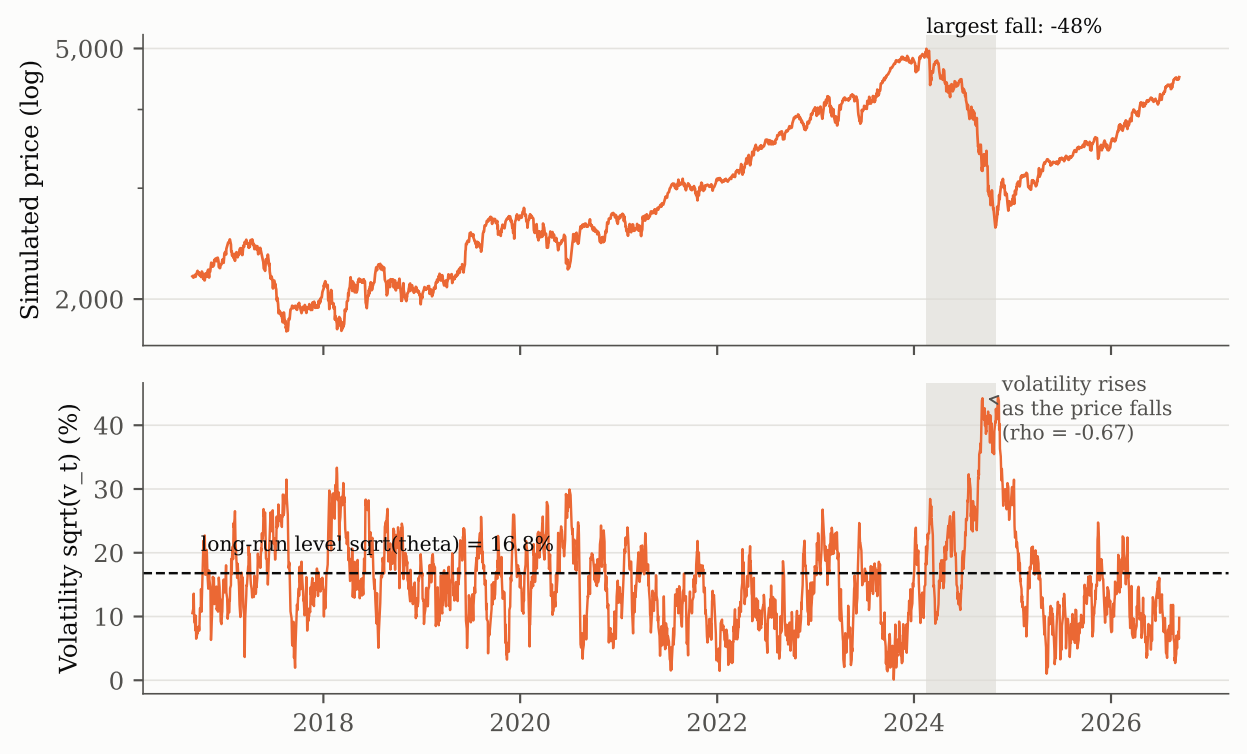
2.3 Heston's Model Results

Table 2: Specification diagnostics – Heston Stochastic Volatility (S&P 500, 2016-09-01 to 2026-09-11, N = 2,519)

Statistic	GBM (Part 4)	Heston	Reference	p-value (Heston)
Panel A: Distribution of PIT residuals				
Skewness	−0.678	−0.451	0	
Kurtosis	19.69	3.704	3	
Jarque-Bera	29,431.8	137.4	$\chi^2(2)$	<0.001
Panel B: Serial dependence (Ljung-Box)				
$Q_{\text{raw}}(10)$	255.7	18.1	$\chi^2(10)$	0.053
$Q_{\text{raw}}(100)$	427.1	112.6	$\chi^2(100)$	0.183
$Q_{\text{sq}}(10)$	2,741.2	71.5	$\chi^2(10)$	<0.001
$Q_{\text{sq}}(100)$	3,281.7	207.5	$\chi^2(100)$	<0.001
Panel C: Variance ratios				
VR(2), cumulated returns / z	0.856	0.962	1	0.073
VR(20), cumulated returns / z	0.806	0.870	1	0.217
VR(10), realized vol 21d	2.579	1.068	1	0.207 (MC)
VR(21), realized vol 63d	5.458	1.360	1	0.010 (MC)
Panel D: Level statistics, model-implied vs. S&P 500 (1,000 simulated paths)				
ADF, log price	−0.86	−1.03 [−2.46, 0.63]	−0.48 (S&P)	0.604 (MC)
ADF, realized vol 21d	−6.79	−5.22 [−6.18, −4.21]	−5.08 (S&P)	0.812 (MC)
OU half-life, RV 21d (days)	13.2	57.8 [45.5, 73.7]	100.5 (S&P)	<0.001 (MC)
OU half-life, RV 63d (days)	40.8	261.8 [173.6, 420.7]	428.0 (S&P)	0.100 (MC)
Panel E: Fit				
Log-likelihood	7,692.8	8,455.2		
AIC	−15,381.7	−16,900.4	lower is better	
BIC	−15,370.0	−16,871.2	lower is better	
Parameters (k)	2	5		

Notes: 'GBM (Part 4)' repeats the Section 4 statistic (raw returns are GBM's PIT residuals); 'Heston' is the same statistic on this model's PIT residuals $z_t = \Phi^{-1}(F_t(r_t))$, i.i.d. $N(0,1)$ under correct specification; 'Reference' is the value a correct model should produce. p-values refer to the Heston column: asymptotic, except (MC) = Monte Carlo (Panel C realized-vol rows against 2,000 i.i.d. $N(0,1)$ series; Panel D: S&P value against 1,000 paths simulated from the fitted model). Variance-ratio p-values use the heteroskedasticity-robust z^*. Panel D: GBM and model columns are simulated medians, model 90% band in brackets; OU p-values are computed on the AR(1) coefficient b. Estimates (annualised, under $\mathbb{P}$): $\mu = 0.1054$, $\kappa = 9.05$, $\theta = 0.02821$, $\xi = 0.63$, $\rho = -0.669$.

Stochastic volatility repairs most of the dynamics: $Q_{sq}(10)$ falls from 2,741.2 to 71.5, raw-return dependence is no longer significant ($p = 0.053$), and short-window volatility persistence is fully explained ($p_{MC} = 0.207$). What remains is negative skew (−0.45; JB = 137.4) and too-fast mean reversion (half-life 58 days against 100). Heston has the best BIC of all models.



2.4 Variance-Gamma Distribution Results

Results mirror Merton: tails are largely fixed (kurtosis 3.95) but each day's activity is independent of the last, so clustering is untouched ($Q_{sq}(10) = 3{,}348.0$).

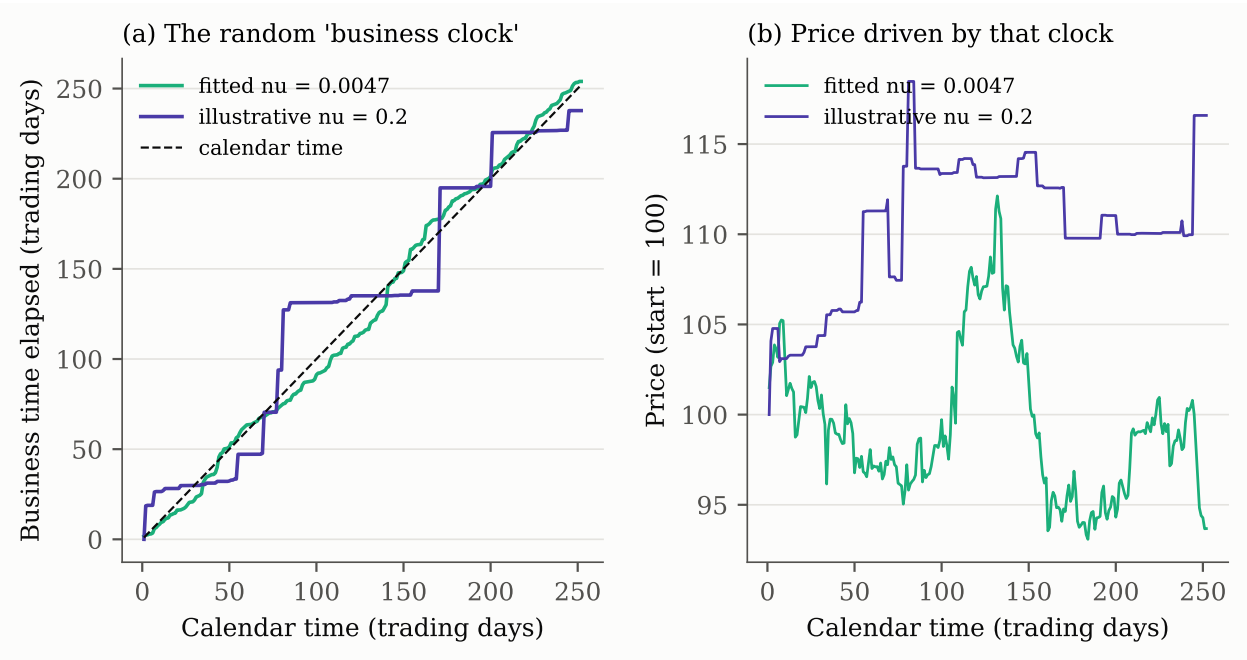


Table 3: Specification diagnostics – Variance Gamma (S&P 500, 2016-09-01 to 2026-09-11, $N = 2{,}519$)

Statistic	GBM (Part 4)	VG	Reference	p-value (VG)
Panel A: Distribution of PIT residuals				
Skewness	−0.678	−0.127	0	
Kurtosis	19.69	3.954	3	
Jarque-Bera	29,431.8	102.3	$\chi^2(2)$	<0.001
Panel B: Serial dependence (Ljung-Box)				
$Q_{\text{raw}}(10)$	255.7	63.8	$\chi^2(10)$	<0.001
$Q_{\text{raw}}(100)$	427.1	191.1	$\chi^2(100)$	<0.001
$Q_{\text{sq}}(10)$	2,741.2	3,348.0	$\chi^2(10)$	<0.001
$Q_{\text{sq}}(100)$	3,281.7	5,288.5	$\chi^2(100)$	<0.001
Panel C: Variance ratios				
VR(2), cumulated returns / z	0.856	0.929	1	0.016
VR(20), cumulated returns / z	0.806	0.774	1	0.113
VR(10), realized vol 21d	2.579	2.284	1	<0.001 (MC)
VR(21), realized vol 63d	5.458	5.716	1	<0.001 (MC)
Panel D: Level statistics, model-implied vs. S&P 500 (1,000 simulated paths)				
ADF, log price	−0.86	−0.86 [−2.46, 0.80]	−0.48 (S&P)	0.690 (MC)
ADF, realized vol 21d	−6.79	−6.74 [−7.72, −5.93]	−5.08 (S&P)	0.002 (MC)
OU half-life, RV 21d (days)	13.2	13.5 [11.5, 15.9]	100.5 (S&P)	<0.001 (MC)
OU half-life, RV 63d (days)	40.8	40.6 [30.4, 53.8]	428.0 (S&P)	<0.001 (MC)
Panel E: Fit				
Log-likelihood	7,692.8	8,127.5		
AIC	−15,381.7	−16,247.1	lower is better	
BIC	−15,370.0	−16,223.8	lower is better	
Parameters (k)	2	4		

Notes: 'GBM (Part 4)' repeats the Section 4 statistic (raw returns are GBM's PIT residuals); 'VG' is the same statistic on this model's PIT residuals $z_t = \Phi^{-1}(F_t(r_t))$, i.i.d. $N(0,1)$ under correct specification; 'Reference' is the value a correct model should produce. p-values refer to the VG column: asymptotic, except (MC) = Monte Carlo (Panel C realized-vol rows against 2,000 i.i.d. $N(0,1)$ series; Panel D: S&P value against 1,000 paths simulated from the fitted model). Variance-ratio p-values use the heteroskedasticity-robust z^*. Panel D: GBM and model columns are simulated medians, model 90% band in brackets; OU p-values are computed on the AR(1) coefficient b. Estimates (annualised, under $\mathbb{P}$): $m = 0.2541$, $\sigma = 0.1675$, $\theta = -0.1277$, $\nu = 0.00469$.

2.5 Bate's Model Results

Adding jumps to Heston improves the tails (JB 137.4 → 84.9, skewness −0.45 → −0.37) but leaves more clustering ($Q_{sq}(10) = 138.2$): the jumps absorb the largest days, so the variance process reacts less. The gain of 8.5 in log-likelihood favors Bates under AIC [Aka74] but not BIC [Sch78], and the jump component is weakly identified — a fit with roughly seven smaller jumps per year performs almost as well.

Table 4: Specification diagnostics – Bates (Stochastic Volatility with Jumps) (S&P 500, 2016-09-01 to 2026-09-11, $N = 2{,}519$)

Statistic	GBM (Part 4)	Bates	Reference	p-value (Bates)
Panel A: Distribution of PIT residuals				
Skewness	−0.678	−0.367	0	
Kurtosis	19.69	3.521	3	
Jarque-Bera	29,431.8	84.9	$\chi^2(2)$	<0.001
Panel B: Serial dependence (Ljung-Box)				
$Q_{\text{raw}}(10)$	255.7	20.5	$\chi^2(10)$	0.025
$Q_{\text{raw}}(100)$	427.1	116.8	$\chi^2(100)$	0.121
$Q_{\text{sq}}(10)$	2,741.2	138.2	$\chi^2(10)$	<0.001
$Q_{\text{sq}}(100)$	3,281.7	280.4	$\chi^2(100)$	<0.001
Panel C: Variance ratios				
VR(2), cumulated returns / z	0.856	0.959	1	0.059
VR(20), cumulated returns / z	0.806	0.871	1	0.231
VR(10), realized vol 21d	2.579	1.225	1	0.001 (MC)
VR(21), realized vol 63d	5.458	1.649	1	<0.001 (MC)
Panel D: Level statistics, model-implied vs. S&P 500 (1,000 simulated paths)				
ADF, log price	−0.86	−1.34 [−2.65, 0.31]	−0.48 (S&P)	0.388 (MC)
ADF, realized vol 21d	−6.79	−4.96 [−6.01, −4.02]	−5.08 (S&P)	0.826 (MC)
OU half-life, RV 21d (days)	13.2	58.1 [43.4, 75.6]	100.5 (S&P)	<0.001 (MC)
OU half-life, RV 63d (days)	40.8	279.1 [179.0, 449.3]	428.0 (S&P)	0.136 (MC)
Panel E: Fit				
Log-likelihood	7,692.8	8,463.7		
AIC	−15,381.7	−16,911.4	lower is better	
BIC	−15,370.0	−16,864.7	lower is better	
Parameters (k)	2	8		

Notes: 'GBM (Part 4)' repeats the Section 4 statistic (raw returns are GBM's PIT residuals); 'Bates' is the same statistic on this model's PIT residuals $z_t = \Phi^{-1}(F_t(r_t))$, i.i.d. $N(0,1)$ under correct specification; 'Reference' is the value a correct model should produce. p-values refer to the Bates column: asymptotic, except (MC) = Monte Carlo (Panel C realized-vol rows against 2,000 i.i.d. $N(0,1)$ series; Panel D: S&P value against 1,000 paths simulated from the fitted model). Variance-ratio p-values use the heteroskedasticity-robust z^*. Panel D: GBM and model columns are simulated medians, model 90% band in brackets; OU p-values are computed on the AR(1) coefficient b. Estimates (annualised, under $\mathbb{P}$): $\mu = 0.0694$, $\kappa = 7.472$, $\theta = 0.02894$, $\xi = 0.5777$, $\rho = -0.7214$, $\lambda = 3.255$, $\mu_J = -0.0171$, $\sigma_J = 0.0009119$. σ_J is at its lower bound (jump size effectively fixed); the jump component is weakly identified (a second start with ∼7 smaller jumps per year fit almost as well).

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